MAIA

Monitoring Assistant & Intelligent Automation

Intelligent Monitoring & Auto-Healing Platform

Executive Overview

Version 2.2 · July 2026

1 What is MAIA?

MAIA — **Monitoring Assistant & Intelligent Automation** — is a rule-based intelligent automation platform that monitors your data pipelines 24/7, detects failures instantly, classifies them automatically, and either fixes them without human intervention or tells your operators exactly what to do — all through a secure, modern web dashboard.

MAIA monitors any job type — DTSX packages, executable processes, Windows services, or any system whose errors can be scanned from log files, database tables, API endpoints, or structured file content. It ships with built-in authentication (three-tier roles), an immutable audit trail, and works fully offline.

Faster Detection	Less Manual Work	Full Audit Trail	Secure & Self-Service
4 scan types detect failures in seconds, not hours	Auto-heal + composite fixes eliminate repetitive work	Every action logged immutably — compliance-ready	Role-gated web UI — no database access needed

2 How It Works

MAIA watches your pipelines, understands what went wrong, knows how to fix it, and either fixes it automatically or tells the operator exactly what to do.

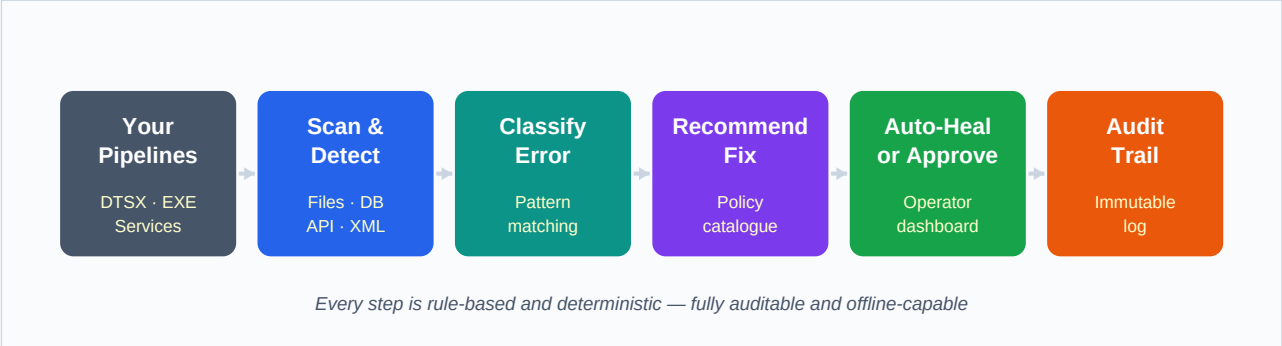

Figure 1 — MAIA Pipeline: From Detection to Resolution

Step	What Happens
1. Scan & Detect	4 strategies monitor log files, database tables, API endpoints, and XML content. Watermarks prevent re-processing.
2. Classify	Configurable pattern rules assign error types. Priority-ordered; first match wins.
3. Recommend Fix	Policy catalogue generates a fix suggestion. Supports single-action and multi-step composite fixes.
4. Auto-Heal or Approve	Known issues resolve automatically. New issues route to operator for one-click approval.
5. Audit Everything	Every detection, classification, fix execution, and operator decision is recorded immutably.

Every step is rule-based and deterministic — fully auditable, offline-capable, and predictable in production.

3 Platform Capabilities

Capability	Description
Multi-Source Monitoring	4 scan strategies (File, Database, API, FileContent). Tier 2.5 ScanSource model: each job has N typed sources with independent rules and watermarks.
Intelligent Classification	Wildcard pattern rules classify every failure. Priority-ordered with per-job overrides. Same matcher across all scan types.
Automated Fix Execution	5 executor types: API calls, stored procedures, SQL scripts, file copy, and manual routing. Composite policies run ordered multi-step fixes.
Auto-Heal	Known issues resolve automatically. Configurable per error type. Every auto-heal is logged. Atomic recommendation claim prevents double-execution.
Operator Dashboard	Modern Angular UI: KPI tiles, analytics charts, failure drawer with keyboard nav, execution history, dark mode, command palette (Ctrl+K).
Security & Audit	PBKDF2 authentication, 3-tier roles (User/Operator/Admin), forced password rotation, immutable audit log with admin viewer.

4 Business Impact

Before MAIA	With MAIA
✕ Manual log scanning every morning	✓ Continuous automated monitoring 24/7
✕ Hours to identify root cause	✓ Failures detected and classified in seconds
✕ Same fixes repeated daily by operators	✓ Known issues auto-healed without intervention
✕ Fix history tracked in spreadsheets	✓ Every action in immutable audit log
✕ No visibility into overnight failures	✓ Real-time dashboard with KPI tiles and charts
✕ Dependent on individual operator knowledge	✓ Knowledge encoded in configurable rules
✕ No access control on operations	✓ 3-tier role-based authentication

90%	24/7	100%	<5 min
Less manual log review	Continuous monitoring	Audit trail coverage	Operator shift review time

5 Roadmap — Current & Next Generation

Today — v2.2 (Built & Operational)

- ✓ Rule-based classification engine with wildcard patterns
- ✓ 4 scan strategies (File, Database, API, FileContent/XML)
- ✓ 5 fix executor types + composite multi-step policies
- ✓ Auto-heal with atomic recommendation claim
- ✓ Full operator dashboard: KPI tiles, charts, drawer, keyboard nav, dark mode
- ✓ Authentication v1: PBKDF2 sessions, 3-tier roles, forced password rotation
- ✓ Immutable audit trail with admin viewer
- ✓ Multi-instance worker (lease-coordinated)
- ✓ Tier 2.5 ScanSource entity with per-job config screen
- ✓ Job Flow view, command palette, server-side sorting

Next Generation — Planned

Capability	Description
LLM-Assisted Classification	Classify novel errors that don't match existing rules using a local language model
Dynamic Fix Suggestions	Language model generates fix candidates for unconfigured error types
ML Anomaly Detection	Detect subtle failure patterns beyond threshold rules
Active Directory SSO	Enterprise authentication alongside MAIA-local auth
Full RTL/Hebrew Support	Complete translations and layout mirroring

The architecture is designed for LLM/ML extension — IClassificationStrategy, IFixCatalogue, and IScanStrategy are the natural injection points. The command palette is built forward-compatible with the planned LLM assistant.

6 Summary

→ MAIA is operational today — all core features built, tested (275+ tests), and deployed

→ Rule-based automation: predictable, auditable, and works fully offline

→ Monitors any job type — DTSX, EXE, services, XML, and more via 4 scan strategies

→ Auto-heal + composite fixes eliminate repetitive manual work for known issues

→ Full authentication with forced password rotation — security built in, not bolted on

→ Immutable audit trail satisfies compliance — every action recorded with actor identity

→ Architecture ready for LLM/ML upgrade when the time is right

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